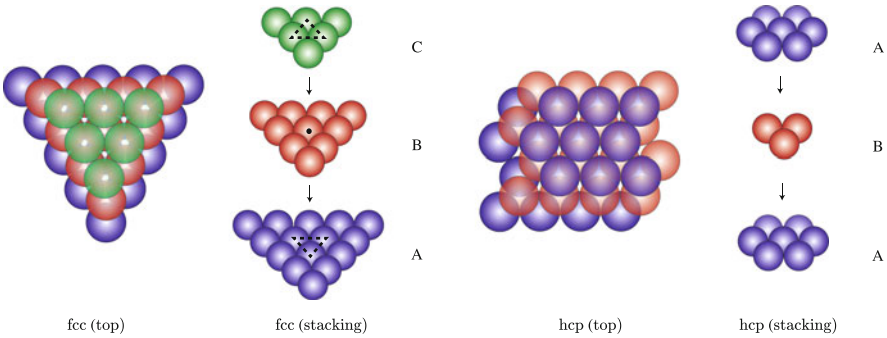



**Fig. 1.18** The face centered cubic and hexagonally close packed lattice are associated to two different stacking arrangements of hexagonally structured planar lattices. The *dashed triangles* indicated in the upper and lower layers of the fcc lattice are centered on the atom indicated by a *dot* in the middle layer

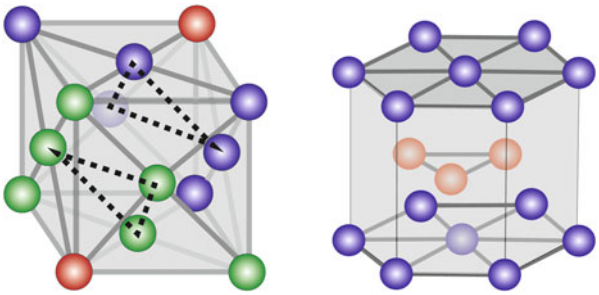


**Fig. 1.19** Unit cells for fcc (*center*), and hcp (*right*) geometries. The coloring of atoms and *dashed triangles* are indicated to match the stacking of layers in Fig. 1.18

(hcp) lattices. The fcc lattice corresponds to the common arrangement by which cannonballs are stacked into pyramidal structures; it can be viewed as a periodic stacking (ABCABC...) of three hexagonally structured planar layers, as illustrated in Fig. 1.18. Also shown in Fig. 1.18 is the hcp lattice, which, on the other hand, alternates two distinct planar lattices. Layer stackings then define a certain unit cell as shown in Fig. 1.19.

### *1.6.1 Lattice Vibrations*

Regardless of the choice of boundary and/or the inclusion of non-pairwise potentials, the minimum of the potential energy occurs where

$$\nabla U = \mathbf{0},$$